PORTOFOLIO | DATA MINING & MACHINE LEARNING

Predictive Modeling for Chronic Kidney Disease (CKD)

Performance and Interpretability Comparison of LASSO, Decision Tree, and Neural Network

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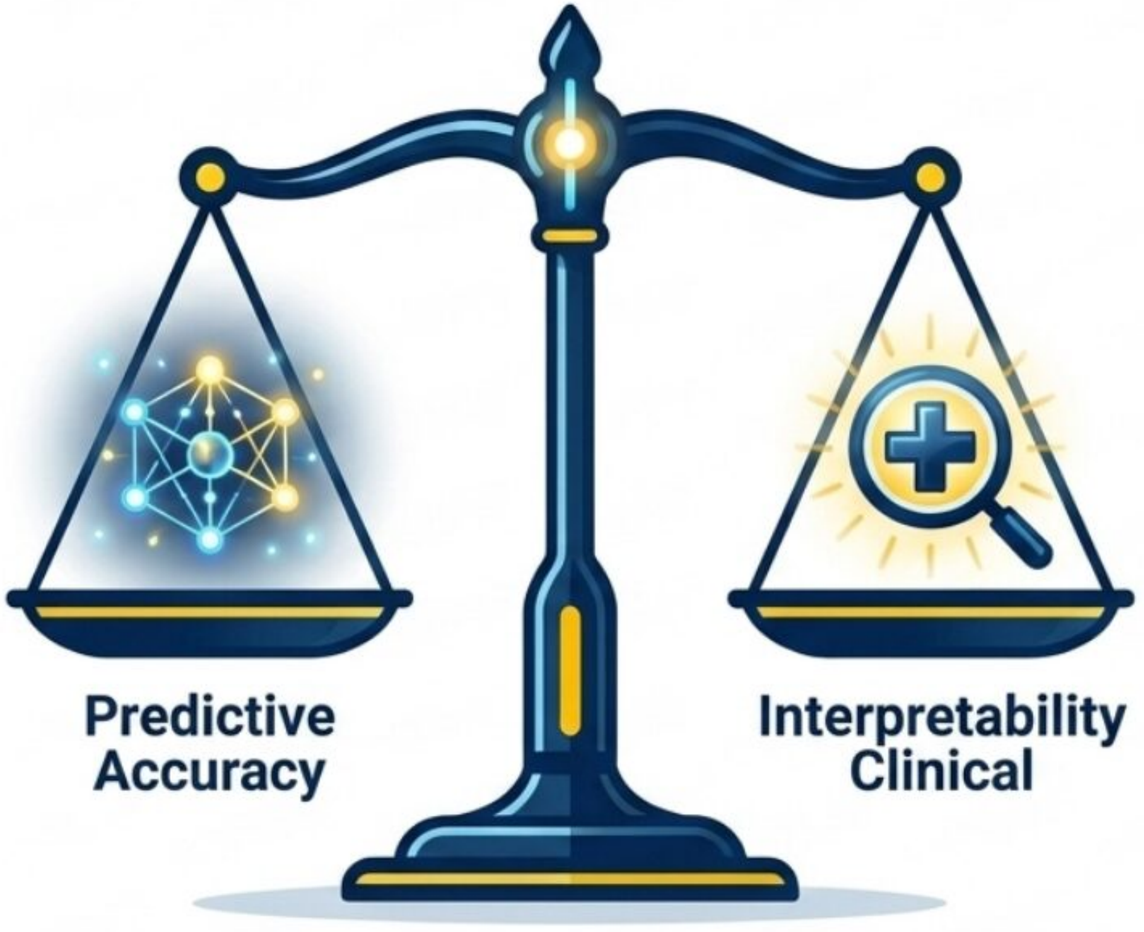
Paradox Modelling Clinics: Accuracy vs. Transparancy



Latar Belakang

Diagnosis CKD mengandalkan kombinasi indikator fungsi ginjal, blood parameters, and komorbidities.

Thallenge

Highly accurate algorithms aractiuated (such as Neural Networks) are often Black-Box, while clinical clinical decisions require transparency (White-Box).

Goal Analysis

Evaluate 3 algorithms (LASSO Logistic Regression, Decision Tree, Neural Network) to find the optimal balance between discrimination ability, calibration, and ease of clinical interpretation klinic.

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Conclusion & Clinical Implementation Roadmap

Main Conclusion

The clinical and laboratory characteristics of the CKD dataset have very strong predictive signals. LASSO Logistic Regression has proven to be the most balanced algorithm for mapping these signals.

Current Limitations

Perfect performance (100% Accuracy) is influenced by the limited sample size in the internal validation set and clear class separation.

Critical Recommendation: External Validation

Before implementation in a hospital Clinical Decision Support System (CDSS), the model **MUST** be tested on datasets from different populations, times, or demographics to ensure real-world parameter stability.

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